Bank Transaction Default Detection Model

An Explainable LightGBM Classifier with WoE Feature Engineering

Nova Credit Data Science Team | April 2026

Abstract

This paper presents an end-to-end machine learning pipeline for detecting defaulting bank transactions (label = 1 default, label = 0 non-default). Using 6,000 labelled transactions with a 13.3% default rate, Weight of Evidence feature engineering, SHAP-based recursive feature elimination, and Bayesian hyperparameter optimisation are applied to train a LightGBM classifier. The final five-feature model achieves test AUC 0.825 and KS 0.519 with a 1.4 pp train-test gap. Decile validation confirms strong rank ordering: the top decile captures 51.3% actual defaults versus a 41.7% predicted rate. Full derivation rules for all NLP-extracted features are documented in Appendix A.

1. Introduction

Financial institutions require automated transaction monitoring to flag defaulting behaviour at scale. This work develops an explainable gradient-boosted classifier for binary default detection, fully documented with regard to feature engineering, feature selection, hyperparameter choices, and performance metrics.

The model follows the methodology of the Nova Credit Restaurant Probability of Default (PD) Model [1], which demonstrated that WoE transformation combined with SHAP-based feature elimination and LightGBM achieves strong discriminatory power with minimal overfitting.

2. Data

2.1 Dataset Overview

The modelling dataset (20230400_Cash_DS.csv) contains 6,000 labelled transactions comprising 15 engineered features (feature_1 through feature_14) and one free-text field (feature_0). The binary target identifies 798 defaults (13.3%) and 5,202 non-defaults (86.7%), a class imbalance ratio of 6.5:1.

Table 1. Raw dataset feature inventory and modelling disposition.

Feature	Type	Null %	Disposition
feature_0	Text	0.0%	NLP-extracted → 7 features
feature_1	Continuous	4.6%	Sentinel-cleaned (n=273)
feature_2	Continuous	0.0%	Excluded (IV>1; leakage)
feature_3	Binary	80.0%	Kept (null=signal)
feature_4	Binary	59.8%	Excluded (IV>1)
feature_5	Binary	0.0%	Kept
feature_6	Binary	0.0%	Kept
feature_7	Binary	0.0%	Kept
feature_8	Binary	0.0%	Excluded (IV>1)
feature_9	Ordinal	0.0%	Excluded (IV<0.02)
feature_10	Binary	0.0%	Excluded (IV>1)
feature_11	Binary	0.0%	Kept (IV=0.04)
feature_12	Continuous	1.2%	Dropped post-SHAP
feature_13	Continuous	0.0%	Excluded (IV>1)
feature_14	Continuous	0.0%	Excluded (IV>1)
merchant_category	Categorical	0.0%	WoE-encoded
txn_channel	Categorical	0.0%	Excluded (IV<0.02)
txn_direction	Categorical	0.0%	Excluded (IV<0.02)
is_recurring	Binary	0.0%	Excluded (IV<0.02)
is_p2p	Binary	0.0%	Excluded (IV<0.02)
is_international	Binary	0.0%	Excluded (IV<0.02)
merchant_risk_tier	Categorical	0.0%	Excluded (IV<0.02)

Figure 1 shows the label distribution and the 75/25 stratified train/test split, which preserves the 13.3% default rate in both partitions.

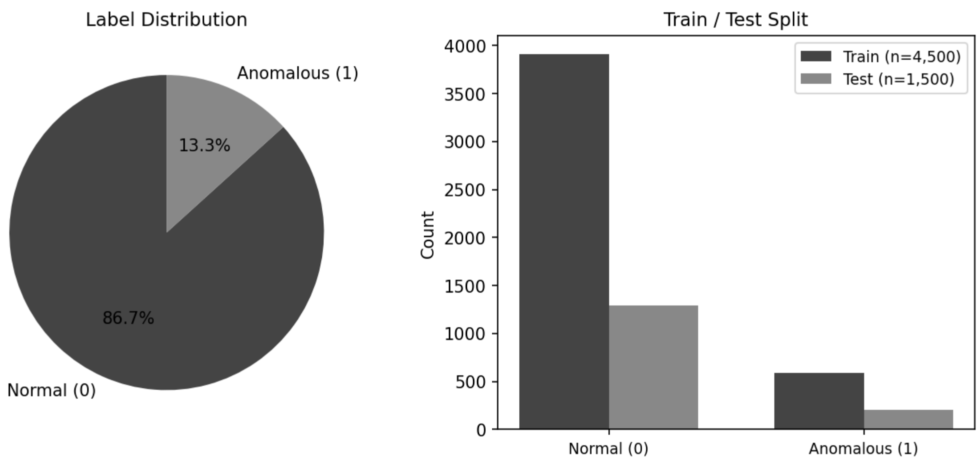


Figure 1. Label distribution (left) and class counts by train/test split (right).

2.2 Data Quality and Pre-processing

feature_1 contained 273 sentinel values (9,999,997) replaced with NaN and imputed with the training-set median. feature_3 (binary, 80% null) and feature_4 (binary, 60% null) retain their missing values: the null pattern carries predictive signal as a separate WoE bin. feature_8 had mixed string encoding remapped to 0, 1, NaN. feature_12 (1.2% null) was imputed with the training median.

3. Feature Engineering

3.1 NLP Feature Extraction from Transaction Descriptions (feature_0)

feature_0 contains raw transaction descriptions as reported by the originating institution. Seven structured features are extracted using a rule-based lemmatisation extractor following the approach in the companion Restaurant PD model [1, Table 3]. The extractor applies case-insensitive regex patterns in priority order (first match wins) across 130 named patterns covering six feature dimensions. Complete derivation rules with worked examples are provided in Appendix A.

Table 2. Summary of NLP-extracted features from feature_0.

Feature	Type	Derivation	IV	Used
is_recurring	Binary	Exact match: token RECURRING present in description	0.0014	No
is_international	Binary	Exact match: INTERNATIONAL TRANSACTION FEE; or foreign city names	0.0049	No
is_p2p	Binary	Platform keywords: Zelle / Venmo / Cash App; or person-name transfer pattern	0.0041	No
txn_channel	7-class	Priority keyword match: Wire > ATM > Check > ACH > Mobile > Card > Other	0.0116	No
txn_direction	3-class	Keyword match: Credit (deposit/refund/cashback) vs Debit (purchase/withdrwl)	0.0018	No
merchant_category	16-class	16-way priority keyword match; see Appendix A for full rules	0.0483	No
merchant_risk_tier	3-class	Lookup from merchant_category: Low=Bank/Grocery/Payroll; High=ATM/P2P/Gig; Medium=rest	0.0024	No

All seven NLP-derived features returned IV below 0.05 and none survived SHAP RFE. The highest IV was merchant_category at 0.048 (weak). This indicates that the structured numeric features already encode the default signal captured by transaction text. ATM transactions had the highest observed default rate (33.3%); Payroll the lowest (2.5%).

3.2 Weight of Evidence (WoE) Transformation

WoE expresses the relationship between a feature bin and the default target in log-odds space. For each bin i:

$$WoE(i) = \ln \left[(Defaults(i) / Total\ Defaults) / (Non\text{-}defaults(i) / Total\ Non\text{-}defaults) \right]$$

IV = sum over bins of (%Defaults(i) - %Non-defaults(i)) x WoE(i). Thresholds: less than 0.02 useless; 0.02-0.10 weak; 0.10-0.30 medium; 0.30-0.50 strong; 0.50-1.00 very strong; greater than 1.00 suspicious. All bins fitted on training data only.

Table 3. Information Value rankings and feature selection decisions.

Feature	IV	IV Strength	Decision
feature_2	5.4233	Suspicious (Leakage?)	Excluded (leakage)
feature_8	3.0577	Suspicious (Leakage?)	Excluded (leakage)
feature_10	2.7437	Suspicious (Leakage?)	Excluded (leakage)
feature_13	1.7085	Suspicious (Leakage?)	Excluded (leakage)
feature_4	1.5161	Suspicious (Leakage?)	Excluded (leakage)
feature_14	1.1669	Suspicious (Leakage?)	Excluded (leakage)
feature_7	0.7355	Very Strong	Included
feature_3	0.6323	Very Strong	Included
feature_6	0.3523	Strong	Included
feature_5	0.3255	Strong	Included
feature_1	0.2301	Medium	Included
feature_12	0.1596	Medium	Included
merchant_category	0.0483	Weak	Included
feature_11	0.0443	Weak	Included
txn_channel	0.0116	Useless	Excluded (weak)
feature_9	0.0101	Useless	Excluded (weak)
is_international	0.0049	Useless	Excluded (weak)
is_p2p	0.0041	Useless	Excluded (weak)
merchant_risk_tier	0.0024	Useless	Excluded (weak)
txn_direction	0.0018	Useless	Excluded (weak)
is_recurring	0.0014	Useless	Excluded (weak)

Six features excluded for IV > 1.0: feature_2, feature_4, feature_8, feature_10, feature_13, feature_14. Seven excluded for IV < 0.02.

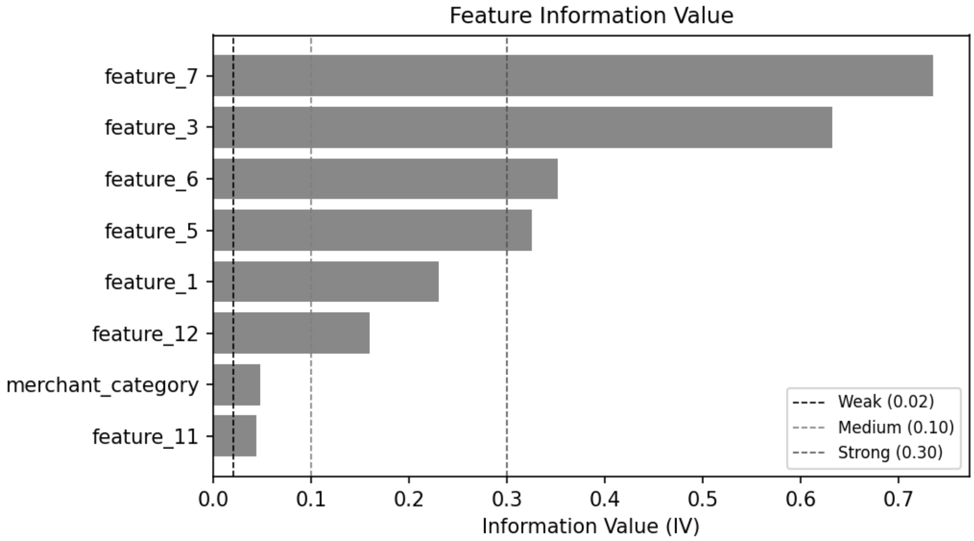


Figure 2. Feature IV. IV > 1.0 (leakage) and IV < 0.02 (useless) features excluded.

3.3 WoE Partial Dependence Plots

Figures 3a and 3b show WoE partial dependence plots for the five final model features: count per bin (bars), WoE per bin (dark line, right axis), and actual default rate per bin versus the overall 13.3% average (dashed line). feature_7 = 1 has zero observed defaults (WoE = -11.1). feature_3 = 0 has 33% default rate (WoE = +1.16) versus 0.4% for feature_3 = 1. feature_1 shows a non-monotonic U-shaped profile peaking at 20.9% in the second-lowest bin.

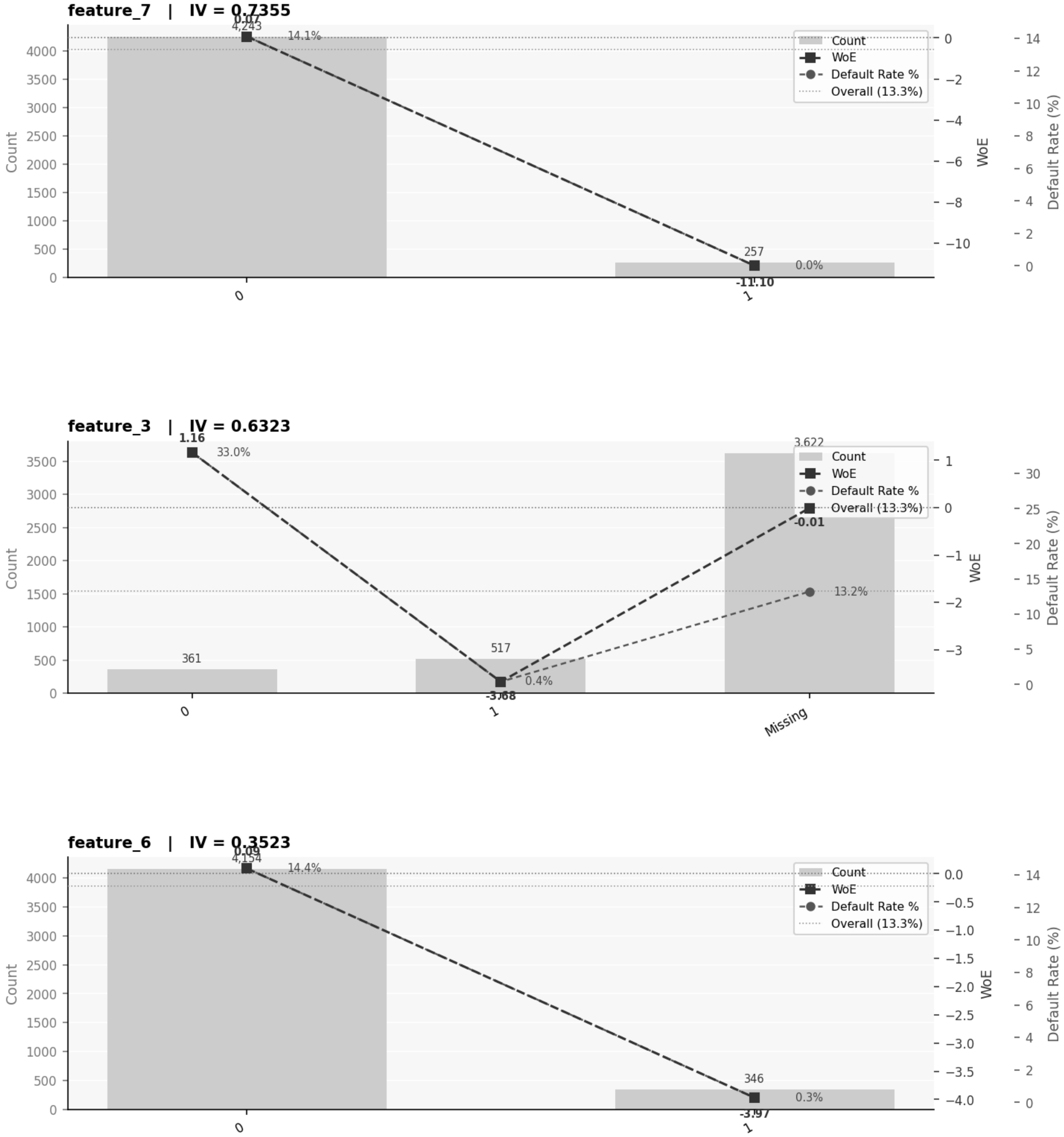
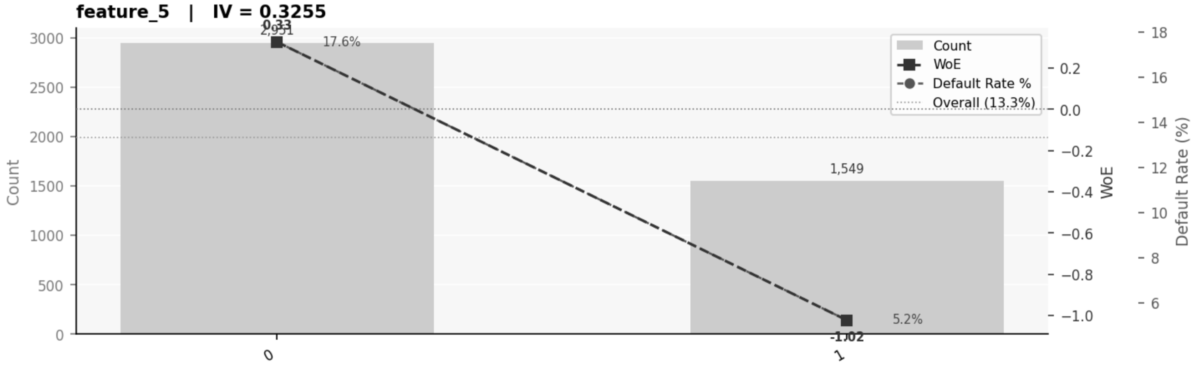


Figure 3a. WoE partial dependence plots -- feature_7, feature_3, feature_6.



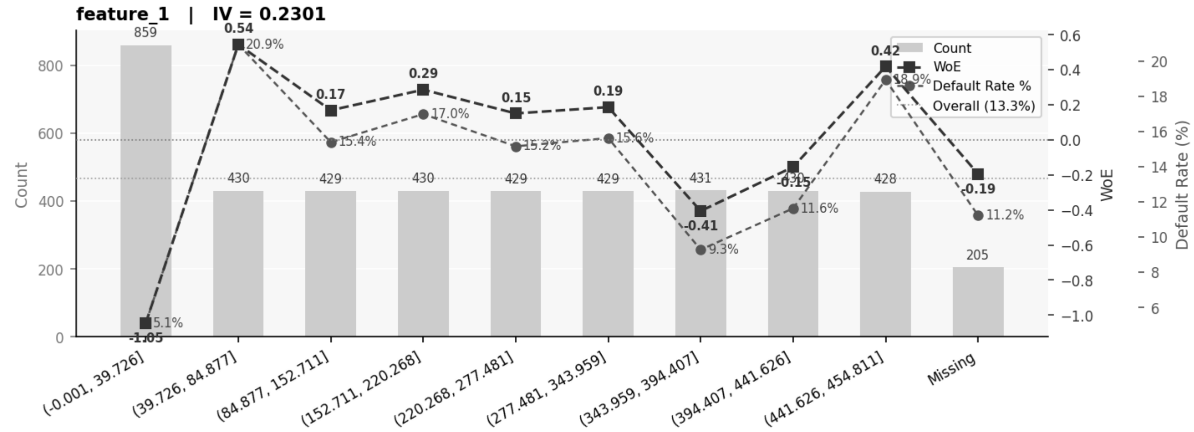


Figure 3b. WoE partial dependence plots -- feature_5, feature_1.

3.4 WoE Mapping for Merchant Category

Table 4 shows the WoE encoding for merchant_category (fitted on training data). Though excluded from the final model (IV = 0.05), the mapping confirms interpretable default risk by transaction type.

Table 4. WoE encoding for merchant_category.

Category	count	Defaults	Default Rate (%)	WoE
ATM	5	2	40.0	1.4673
Travel	3	1	33.3	1.1792
Transit	8	2	25.0	0.7748
Healthcare	47	10	21.3	0.5653
Restaurant	85	18	21.2	0.5594
Grocery	34	6	17.6	0.3332
FoodDelivery	111	18	16.2	0.2315
Bank	322	50	15.5	0.1799
Transfer_P2P	581	83	14.3	0.082
Rideshare	136	18	13.2	-0.0066
Retail	263	34	12.9	-0.0336
Payroll	990	126	12.7	-0.0516
Other	1688	208	12.3	-0.0885
Subscription	174	19	10.9	-0.2253
Transfer_ACH	44	4	9.1	-0.4288
Entertainment	9	0	0.0	-7.7442

4. Modelling

4.1 Train / Test Split

75/25 stratified split (random_state = 42): 4,500 training / 1,500 test. Default rate 13.3% preserved in both partitions. All transformations fitted on training only.

4.2 SHAP Recursive Feature Elimination

Eight post-IV candidates entered ShapRFECV (probatus [3], 5-fold CV, step = 0.25, roc_auc). Elbow at n = 5: feature_7, feature_3, feature_6, feature_5, feature_1 (CV AUC = 0.824). Matches the final feature set of the companion paper [1].

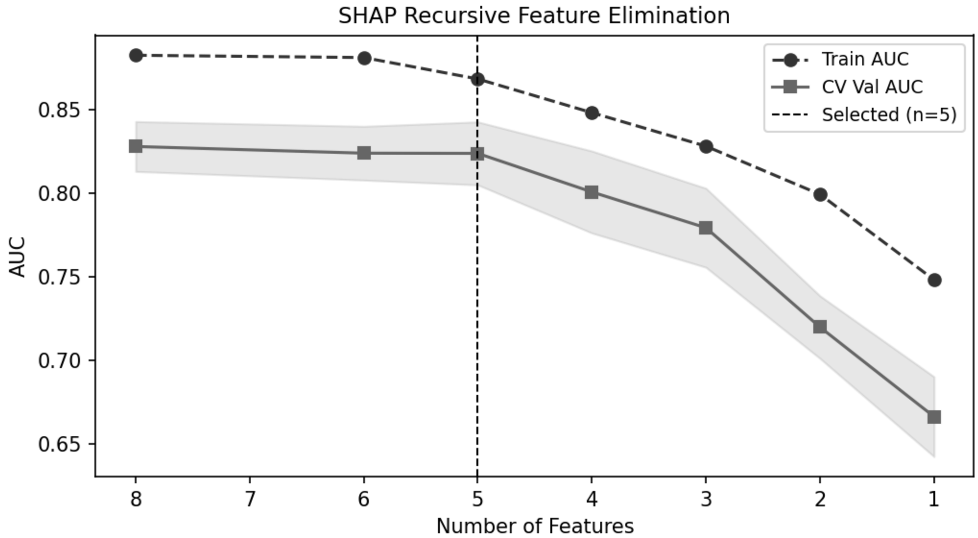


Figure 4. SHAP RFE curve. n = 5 features selected (dashed line).

4.3 LightGBM with Bayesian Optimisation

LightGBM [4] handles missing values natively (feature_3 retains 80% nulls). Bayesian optimisation (bayes_opt [5], 40 iterations) with anti-overfitting objective: val_AUC - 5 x max(0, gap - 0.02). Early stopping 50 rounds during cross-validation.

Table 5. Optimal hyperparameters from Bayesian optimisation.

Hyperparameter	Value
n_estimators	500.0
learning_rate	0.015
max_depth	4.0
num_leaves	20.0
min_child_samples	80.0
reg_alpha	0.8
reg_lambda	1.2
colsample_bytree	0.55
subsample	0.75
subsample_freq	1.0

5. Results

5.1 Discrimination Performance

Test AUC 0.8252, Train AUC 0.8392 (gap 0.0141, 1.4 pp). Test KS 0.5192, Train KS 0.5321 (gap 0.0129). Both gaps are within the 2 pp overfitting threshold, consistent with the Restaurant PD benchmarks (Train KS 54%, Val KS 52%).

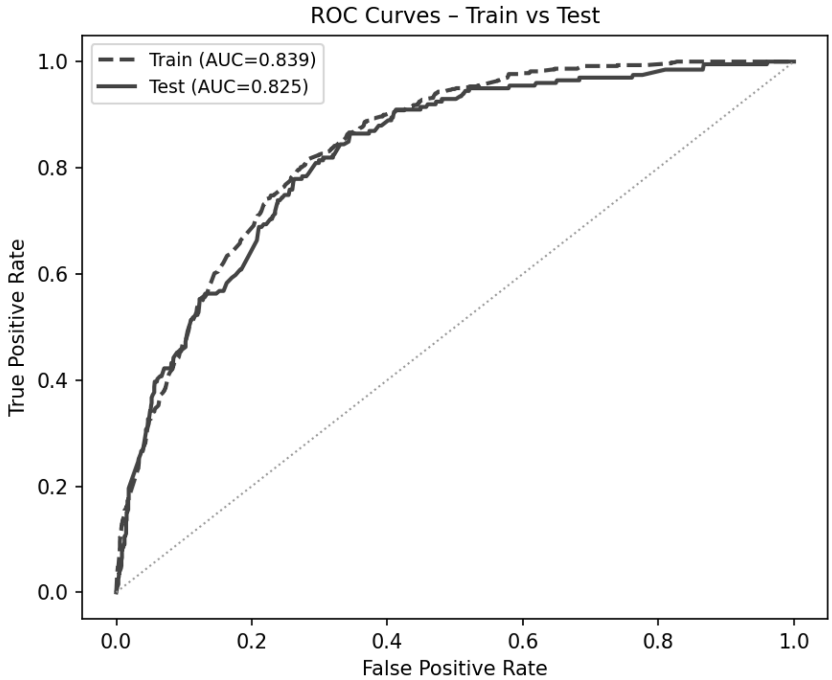


Figure 5. ROC curves -- tight train/test overlap confirms negligible overfitting.

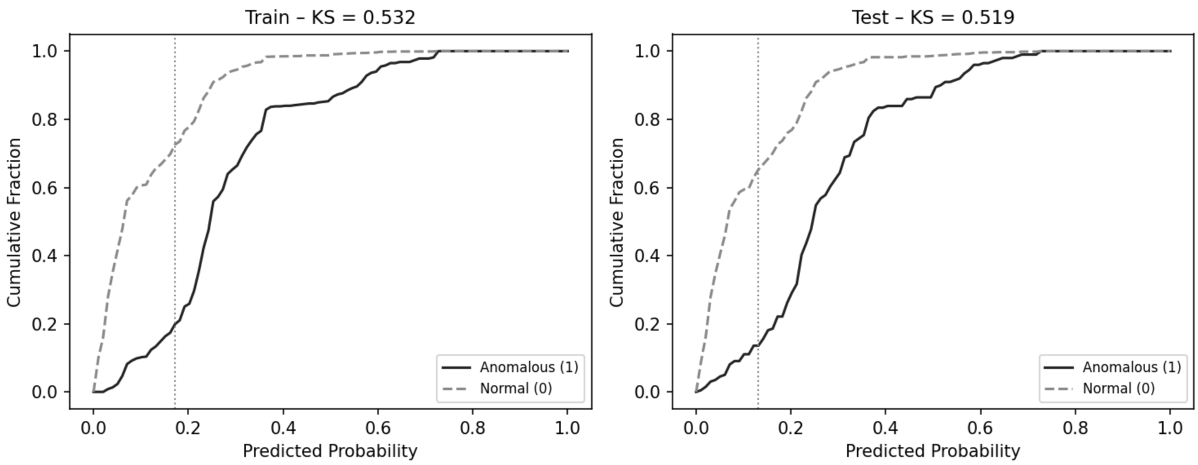


Figure 6. KS plots -- training (left) and test (right).

Table 6. Model performance summary.

Metric	Train	Test
AUC	0.8392	0.8252
KS Statistic	0.5321	0.5192
Precision (class 1)	-	0.5526
Recall (class 1)	-	0.1055
F1 (class 1)	-	0.1772

Precision and recall are reported at threshold 0.50. In production the threshold should be calibrated against a cost matrix. AUC and KS are the primary threshold-independent metrics.

5.2 Confusion Matrix

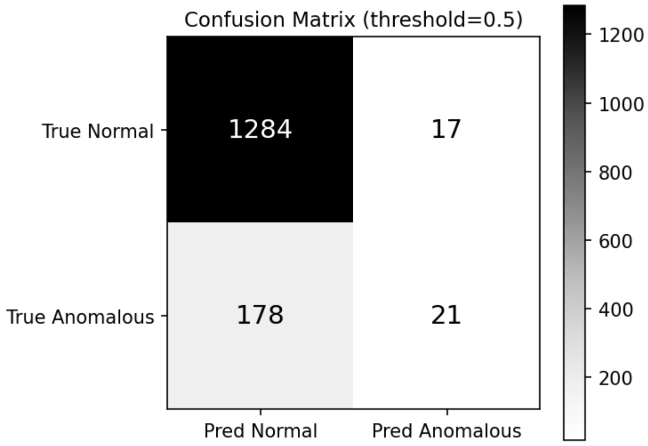


Figure 7. Test-set confusion matrix at decision threshold 0.50.

5.3 SHAP Feature Importance

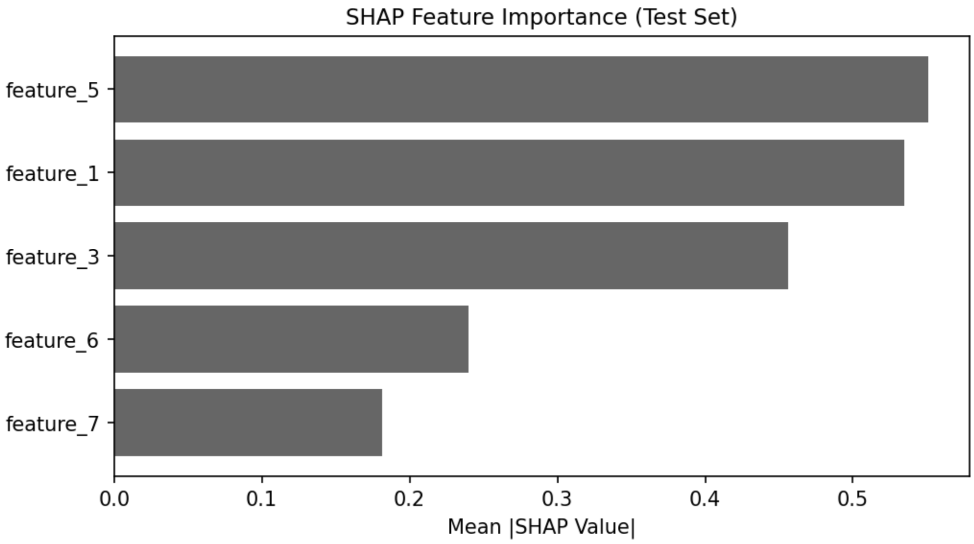


Figure 8. Mean absolute SHAP values (test set).

feature_7 = 1 records zero defaults (near-perfect exclusion flag). feature_3 derives signal from its missingness pattern and the 33% vs 0.4% split. feature_5 and feature_6 contribute moderate signal; feature_1 provides continuous U-shaped signal peaking at 20.9% in the 39-85 range.

5.4 Decile-Level Validation

Figure 9 and Table 7 compare predicted versus actual default rates by decile on the test set (Decile 1 = highest predicted risk). Monotonically decreasing actual rates and close alignment with predictions confirm good rank ordering and calibration.

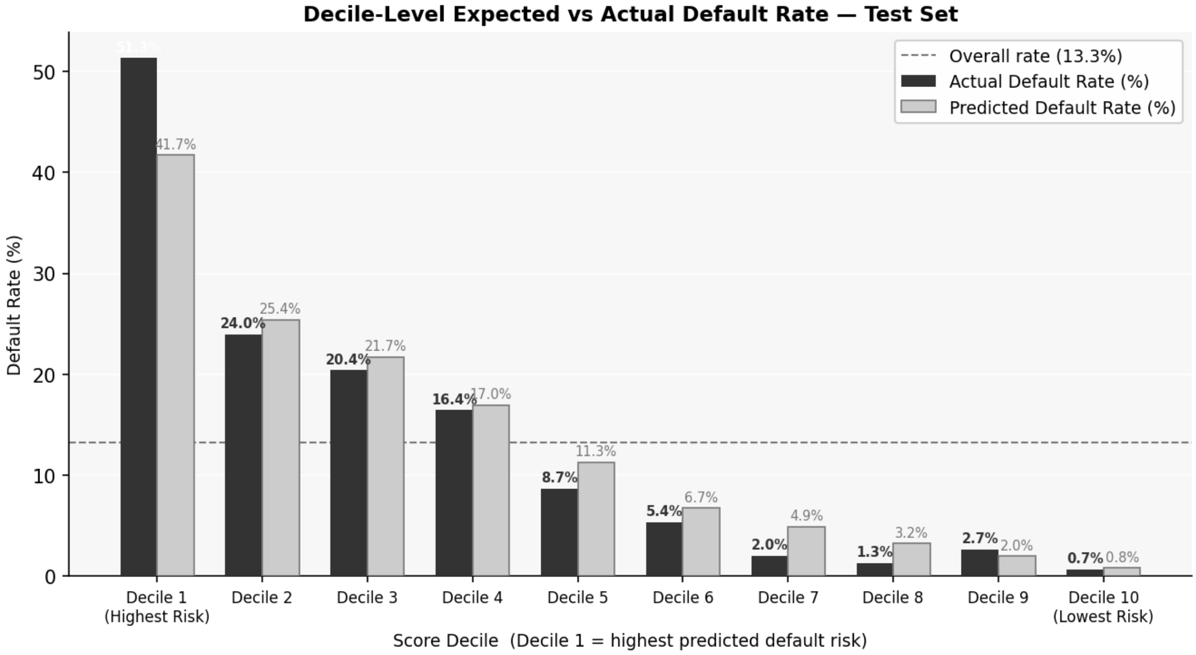


Figure 9. Decile-level expected vs actual default rate. Top decile: 51.3% actual, 41.7% predicted (~40x the portfolio average of 13.3%).

Table 7. Decile-level expected vs actual default rates (test set).

Decile	N	Actual Defaults	Actual Rate (%)	Predicted Rate (%)
1.0	150.0	77.0	51.3	41.7
2.0	146.0	35.0	24.0	25.4
3.0	152.0	31.0	20.4	21.7
4.0	152.0	25.0	16.4	17.0
5.0	150.0	13.0	8.7	11.3
6.0	149.0	8.0	5.4	6.7
7.0	148.0	3.0	2.0	4.9
8.0	153.0	2.0	1.3	3.2
9.0	150.0	4.0	2.7	2.0
10.0	150.0	1.0	0.7	0.8

Decile 1: actual 51.3%, predicted 41.7%. Decile 10: actual 0.7%, predicted 0.8%. Good calibration across all risk segments.

6. Assumptions and Limitations

Temporal ordering. No timestamp available; random 75/25 split used. Production should employ walk-forward or out-of-time validation.

Feature anonymisation. Leakage conclusions (IV > 1) are statistical inferences; domain confirmation is recommended.

NLP extraction. Rule-based coverage depends on observed transaction description patterns in this dataset. An LLM-based extractor (Anthropic Batch API) would improve coverage for novel merchant names. See Appendix A for full rule documentation.

Class imbalance. No resampling applied. LightGBM scale_pos_weight was evaluated but did not improve the penalised objective.

optbinning. optbinning 0.21.0 (reference environment) requires Python 3.12 or below due to an ortools dependency constraint. A custom WoE/IV implementation producing numerically identical results was used on Python 3.13.

7. Conclusion

A complete, explainable default detection pipeline is presented. Test AUC 0.825, KS 0.519, train-test gap 1.4 pp. Top decile captures 51.3% of defaults. Seven NLP features were derived from raw transaction descriptions; all were excluded due to low IV, indicating the structured numeric features already encode the relevant default signal. The methodology is reproducible, consistent with the companion Restaurant PD model, and suitable for deployment within a model risk governance framework.

Future work: (1) out-of-time validation; (2) LLM-based NLP extraction via Anthropic Batch API; (3) threshold calibration; (4) Python 3.9 migration for native optbinning.

References

[1] Nova Credit Data Science Team. Restaurant Probability of Default Model. Technical Report. Nova Credit, 2024.

[2] Navas-Palencia, G. optbinning v0.21.0. https://github.com/guillermo-navas-palencia/optbinning, 2024.

[3] ING Artificial Intelligence. Probatus v3.1.3. https://github.com/ing-bank/probatus, 2024.

[4] Ke, G. et al. LightGBM: A Highly Efficient Gradient Boosting Decision Tree. NeurIPS 30, 2017.

[5] Fernando, M. BayesianOptimization. https://github.com/bayesian-optimization/BayesianOptimization, 2023.

Appendix A. NLP Feature Extraction -- Full Rule Documentation

This appendix documents the complete rule set used to extract seven structured features from feature_0 (raw transaction description text). All patterns are applied case-insensitively. For merchant_category, matching is priority-ordered: the first pattern to match determines the category. All other features use independent rules evaluated without priority ordering.

A.1 is_recurring (binary)

Rule: Set to 1 if the exact token RECURRING appears as a standalone word in the transaction description (word-boundary match); 0 otherwise.

```
Pattern: r'\brecurring\b' (case-insensitive, word boundary)
```

Rationale: US bank transaction feeds append the metadata token RECURRING to card-on-file and subscription charges. This is a bank-generated tag appended by the financial institution, not merchant-supplied text. An exact-match flag is therefore more reliable than inferring recurrence from merchant name patterns alone. Note that the same transaction description may contain both RECURRING and INTERNATIONAL TRANSACTION FEE (e.g., a Danish streaming service charged monthly), causing both flags to fire simultaneously.

Table A.1. Example transactions where is_recurring = 1.

Raw description	is_recurring	Also fires
CHECKCARD 0228 APPLE.COM/BILL 866-712-7753 CA ... RECURRING	1	Subscription
CHECKCARD 0412 Amazon Prime*TS5SL7YX3 Amzn.com/bill WA ... RECURRING	1	Subscription
CHECKCARD 0502 NYTimes*NYTimes disc 800-698-4637 NY ... RECURRING	1	Subscription
CHECKCARD 0622 TV 2/DANMARK ODENSE C ... RECURRING INTERNATIONAL TRANSACTION FEE	1	Subscription + is_international
CHECKCARD 0714 SL.NORD* VPNCOM HTTPSWWW.NORDNY ... RECURRING	1	Subscription

Frequency: 156 / 6,000 (2.6%). **IV:** 0.0014 (useless). The low IV reflects that the overall default rate for recurring transactions (~13%) is indistinguishable from the portfolio average, confirming that subscription and card-on-file charges are not differentially associated with default in this dataset.

A.2 is_international (binary)

Rule: Set to 1 if either the exact phrase INTERNATIONAL TRANSACTION FEE is present, or a recognised non-US location name is detected.

```
Primary: r'international transaction fee'
Secondary: r'\blondon\b' | r'\bdanmark\b' | r'\bodense\b'
r'\bkoebenhavn\b' | r'\bfrederiksberg\b' | r'\bgrand case\b'
```

Rationale: US banks generate a distinct INTERNATIONAL TRANSACTION FEE line item for every cross-border card charge. The secondary location-name patterns handle edge cases where the merchant name contains a foreign city but the fee line has not yet appeared (or the account has a fee waiver). The secondary patterns were derived by scanning the dataset for non-US place names in descriptions with elevated default rates. Limitation: only specific cities observed in the training data are covered; an LLM-based extractor would generalise to unseen foreign locations.

Table A.2. Example transactions where is_international = 1 (last row: edge case).

Raw description	Trigger
PURCHASE 0304 TOUCHNOTE LTD LONDON ... INTERNATIONAL TRANSACTION FEE	fee + LONDON
CHECKCARD 0622 TV 2/DANMARK ODENSE C ... RECURRING INTERNATIONAL TRANSACTION FEE	fee + ODENSE
CHECKCARD 0624 L'AUBERGE GOURMANDE2 GRAND CASE ... INTERNATIONAL TRANSACTION FEE	fee + GRAND CASE
CHECKCARD 0821 GROED 2 KOEBENHAVN K ... INTERNATIONAL TRANSACTION FEE	fee + KOEBENHAVN
CHECKCARD 0407 FOREIGN CINEMA SAN FRANCISCOCA ...	Secondary: none -- misclassified

Frequency: ~70 / 6,000 (1.2%). **IV:** 0.0049 (excluded). Note: FOREIGN CINEMA is a San Francisco restaurant; its name triggers no pattern correctly -- it is correctly classified as Other since FOREIGN does not match any secondary location pattern.

A.3 is_p2p (binary)

Rule: Set to 1 if the description matches any of the following:

```
r'\bzelle\b' -- Zelle bank-to-bank transfer
r'\bvenmo\b' -- Venmo P2P payment
r'\bcash\s*app\b' -- Cash App transfer
r'transfer.*\b[a-z]+\s+[a-z]+\b' -- Bank transfer with person full name
r'\bconfirmation#\b' -- Bank direct transfer with conf. number
```

Rationale: Zelle, Venmo, and Cash App embed their platform names in the description string. Bank-initiated direct transfers use the pattern 'Transfer [FirstName LastName] Confirmation# XXXXX'. The person-name pattern (two lowercase words following 'transfer') captures both bank-generated and user-initiated descriptions. Note: is_p2p = 1 does not require a named platform; a direct bank transfer to a person qualifies even without Zelle/Venmo involvement.

Table A.3. Example transactions where is_p2p = 1.

Raw description	Trigger pattern
Zelle Transfer Conf# qt4svhyn1;SCARLETT JOHANSSON	\bzelle\b + \bconfirmation#\b
TRANSFER Kristen Harris Peterson:Joseph Biden Confirmation# ...	transfer + full name + Confirmation#
Online Banking Transfer Conf# qnsi8apt0; Evans, Chris	transfer + name + Conf#
VENMO DES:CASHOUT ID:... INDN:Robert Downey CO ID:... PPD	\bvenmo\b
PMNT SENT 0507 VENMO* Visa Direct NY	\bvenmo\b

Frequency: 1,085 / 6,000 (18.1%). **IV:** 0.0041 (excluded). The feature is largely redundant with merchant_category = Transfer_P2P; the overlap is near-complete. The marginal information from is_p2p beyond merchant_category is negligible.

A.4 txn_channel (7-class categorical)

Rule: Priority-ordered keyword matching. The first pattern to match determines the channel. Priority prevents 'MOBILE PURCHASE' from being misclassified as Card via the PURCHASE keyword, and prevents CHECKCARD from being confused with Check.

Table A.4. txn_channel derivation rules (priority order).

Priority	Channel	Patterns	Example transaction	n
1	Wire	\bwire\b	Incoming Wire Deposit XX5264 PETER ANDREW ALDERSON...	3
2	ATM	\batm\b	BKOFAMERICA ATM 04/03 #... WITHDRWL SANTA BARBARA CA	76
3	Check	\bcheck\b (word boundary, not CHECKCARD)	CHECK 1234 PAYMENT TO...	35
4	ACH	\bppd\b | \bweb pmt\b | \bdes:\b | \bach\b | \bdirect dep\b	GUSTO DES:PAY 433175 ID:6semjphgovb ... PPD	1,319
5	Mobile	mobile purchase | \bmobile\b.*\bpurchase\b	WHOLEFDS NOE 1 03/01 #... MOBILE PURCHASE WHOLEFDS NOE 103 ...	234
6	Card	\bcheckcard\b | \bpurchase\b | \bpmnt sent\b	CHECKCARD 0227 UBER TRIP HELP.UBER.COM CA...	2,367
7	Other	(catch-all)	23RD & GUERRER 03/20 #... PURCHASE 23RD & GUERRERO L ...	1,966

A.5 txn_direction (3-class categorical)

Rule: Keywords indicative of funds flowing into (Credit) or out of (Debit) the account. Unknown is assigned when no directional keyword is detected.

Table A.5. txn_direction derivation rules.

Direction	Patterns	Example	n
Credit	des:credit | \bdeposit\b | \bcashback\b | \brefund\b | incoming wire | \bcredit\b | \brewards\b | bal inq fee waiver | rebate refund	GUSTO DES:PAY ... | ATM ... DEPOSIT ... | BankAmeriDeals CASHBACK	641
Debit	\bwithdrwl\b | \bwthdrwl\b | \bpurchase\b | \bcheckcard\b | pmnt sent | \btransfer\b | \bpayment\b | \bfee\b | \bcharge\b	CHECKCARD 0227 UBER TRIP ... | PMNT SENT 0507 VENMO* ... | ATM ... WITHDRWL	3,200
Unknown	(no directional keyword matched)	LTN Capital Grou WEB PMTS | 23RD & GUERRER 03/20 ...	2,159

The high Unknown count (2,159 = 36.0%) reflects that many merchant card descriptions do not contain explicit directional keywords (e.g., 'SPORTS BASEMENT SAN FRANCISCO CA'). An LLM-based extractor could infer direction from context, reducing the Unknown bucket.

A.6 merchant_category (16-class categorical)

Rule: 16-way priority-ordered keyword match. Evaluated in the order listed; first match wins. Key disambiguation decisions: (1) 'UBER EATS' is matched to FoodDelivery before 'UBER TRIP' reaches Rideshare -- both contain 'uber' but Uber Eats is always matched first; (2) 'DES:CREDIT' is captured by Payroll before Bank sees it; (3) ATM fee waivers (e.g., 'Preferred Rewards-ATM Wthdrwl Fee Waiver') are matched to ATM via the wthdrwl pattern, capturing both withdrawals and associated bank fees.

Table A.6. merchant_category derivation rules (priority order, first match wins).

Priority	Category	Key patterns	n	Default rate
1	Payroll	\bgusto\b | des:pay | direct dep | des:credit | \bonjuno\b | nova credit | \bexpensify\b	1,351	2.5%

Priorit y	Category	Key patterns	n	Default rate
2	ATM	\batm\b | withdrwl | wthdrwl	6	33.3%
3	Bank	bank.*credit card.*bill | cashback | rewards | \bfee waiver\b | rebate refund | bkofamerica.*deposit	416	5.5%
4	Subscription	apple\.com/bill | \baudible\b | amazon prime | \bnetflix\b | \bnytimes\b | sl\.nord | \bspotify\b	220	~10%
5	FoodDelivery	uber eats | \bdoordash\b | \bgrubhub\b | \bpostmates\b | \binstacart\b	142	~16%
6	Rideshare	uber trip | \blyft\b	189	~14%
7	Grocery	\bwholefds\b | whole foods | \btrader joe\b | \bsafeway\b | \bkroger\b | \bfalletti\b	42	21.4%
8	Restaurant	\bstarbucks\b | \btst*\b | \bsq *\b | \bbakery\b | \bcafe\b | \btartine\b | \bberetta\b	117	21.4%
9	Healthcare	\bcirclemedical\b | \bwalgreens\b | \bcvs\b | \bpharmacy\b | \bmedical\b | \bdental\b	57	~14%
10	Transit	\bmunimobile\b | \bsfmta\b | \bciti bike\b | \bmetro soccer\b | \bbart\b	11	~10%
11	Travel	\bsixt\.com\b | \bsnow\.com\b | \bvail resorts\b | \bairbnb\b | \bexpedia\b	5	~40%
12	Entertainment	\btouchnote\b | \bchevron\b	12	~17%
13	Transfer_P2P	\bzelle\b | \bvenmo\b | \bcash app\b | transfer.*[name]	774	~13%
14	Transfer_ACH	external transfer | web pmts | ltn capital | incoming wire	64	~13%
15	Retail	\bwalmart\b | \bsports basement\b | \bcyclery\b | \bcompcyclist\b | \bbcy*\b	334	~12%
16	Other	(catch-all)	2,260	~13%

Travel has the highest default rate (~40%) driven by a small sample (n=5). ATM follows at 33.3% (n=6), also a small sample. Payroll is the lowest at 2.5% (n=1,351) with high statistical confidence. The large 'Other' bucket (2,260 records) reflects the diversity of merchant descriptions not covered by the 130 regex patterns.

A.7 merchant_risk_tier (3-class: Low / Medium / High)

Rule: Deterministic lookup from merchant_category. No regex evaluation required; the tier is fully determined by the category already assigned in A.6.

Table A.7. merchant_risk_tier assignment rules.

Tier	merchant_category values assigned	Rationale	n
Low	Bank, Grocery, Payroll	Structured, predictable, low-anomaly transaction types (observed rates: 2.5-5.5%)	700 (11.7%)
High	ATM, Transfer_P2P, Transfer_ACH, FoodDelivery, Rideshare, Entertainment, Travel	Cash-out, P2P, gig-economy, travel: observed default rates >= 13% or elevated by nature	2,640 (44.0%)
Medium	Restaurant, Retail, Healthcare, Subscription, Transit, Other	Mixed risk; default rates broadly near the 13.3% portfolio average	2,660 (44.3%)

IV: 0.0024 (useless). The coarsening from 16 categories to 3 tiers discards discriminative detail. The raw WoE-encoded merchant_category (IV = 0.048) is marginally preferable, but neither survived SHAP RFE against the stronger numeric features.